

Rakesh Sharma

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SUMMARY

Curious data scientist with a robust background in bioengineering and biomedical technologies. Proven expertise in developing innovative imaging algorithms, automating workflows, and spearheading surgical robotic solutions. Adept at leading teams, fostering collaboration, and driving high-stakes projects from conception to execution, all while maintaining a focus on clinical validation and data science for impactful healthcare advancements. [click here](#).

EDUCATION

- 2024 – present PhD in **Bioengineering, University of Pennsylvania** (GPA: 4.0/4.0)
Advisor: Prof. Walter RT Witschey and Prof. Victoria M Gershuni
Research: LLM-agent for medical image analysis; Clinically-aware LLM-as-a-judge; CT imaging -driven post-operative outcome prediction.
- 2008 – 2013 B.Tech and M.Tech. **Bioengineering, IIT (BHU) Varanasi** (GPA: 8.57/10)
Advisor: Prof. Neeraj Sharma and Prof. Nira Misra
Research: Digital signal processing, biomedical image analysis, bio-material synthesis, bio-sensor fabrication..

WORK EXPERIENCE

- Data Scientist, Eli Lilly and Company, Bengaluru, India** 2020 - 2024
- Responsible for building and deploying data science solutions.
 - Product research and development workflow automation.
 - Data science for drug delivery device innovation.
 - Imaging pipeline and algorithm development for semi-automated/automated device failure analysis.
- CTO and Co-founder, Comofi Medtech Pvt Ltd, Bengaluru, India** 2017 - 2020
- Led end-to-end surgical robotic platform development.
 - Developed novel medical image analysis algorithms for segmentation and registration.
 - Managed funding rounds, grant proposals, and manufacturing setup during COVID.
- Scientist, Achira Labs Pvt Ltd, Bengaluru, India** 2015 - 2017
- Microfluidics-based in-vitro diagnostic platform development.
 - Hormone analysis panel development and clinical validation.
 - Mentored junior scientists, fostering their professional development skills.
- Engineer (CTO Office), Wipro Technologies, Bengaluru, India** 2013 - 2015
- Innovative computer vision-based solution development for the financial sector.
 - Product pitch and demonstrations to prospective clients.

PUBLICATIONS

Patent Applications

Desai, Hetalkumar et al. (Mar. 2023). *WO2023044089A1 - Methods and apparatuses for detecting anomalies when filling a container with fluid*. US Patent App. PCT/US2022/043937.

Gururaj, KB et al. (Jan. 2022). *System for renal puncturing assistance*. US Patent App. 17/294,009.

Sharma, Rakesh and Manoj Madhusudhanan (Dec. 2015). *Image processing method of enabling financial transaction and an image processing system thereof*. US Patent App. 14/459,428.

Journal Papers

Beeche, Cameron et al. (2025). “A Pan-Organ Vision-Language Model for Generalizable 3D CT Representations”. In: *medRxiv*.

Dang, Xiangnan et al. (2024). “Clinical investigation of large volume subcutaneous delivery up to 25 mL for lean and non-lean subjects”. In: *Pharmaceutical Research* 41.4, pp. 751–763.

Montoya, Gustavo et al. (2024). “Computational Modeling and Simulation of the Cerebrospinal Flow and Drug Delivery”. In: *Alzheimer’s & Dementia* 20, e094612.

Goyal, Saurabh et al. (2020). “Optimal sample pooling: an efficient tool against SARS-CoV-2”. In: *medRxiv*.

Sharma, Rakesh, Govinda Kapusetti, et al. (2017). “Osteoconductive amine-functionalized graphene–poly (methyl methacrylate) bone cement composite with controlled exothermic polymerization”. In: *Bioconjugate chemistry* 28.9, pp. 2254–2265.

Conference Proceedings

Chen, Hongzhuo et al. (2025). “Adapting Vision-Language Models for 3D CT/MRI Understanding on PMBB via Slice Selection and Explanation Analysis”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 2273–2282.

Pai, Jayeeta et al. (2019). “A Point-of-care immunoassay platform for thyroid function based on hydrogel sensors embedded inside a microfluidic device”. In: *23rd International Conference on Miniaturized Systems for Chemistry and Life Sciences, MicroTAS 2019*. Chemical and Biological Microsystems Society, pp. 102–103.

SKILLS

Tools Python, PyTorch, ITK-SNAP, Slicer3D, SimpleITK, OpenCV

Skills Medical image analysis, Natural Language Processing, Statistical analysis, Project Management, Workflow automation

REFERENCES

Available upon request.